

Claude Ecosystem & Spec-Driven Development

Spec-Kit vs GStack — Deep Comparison

1. The Claude Ecosystem (2026)

CLAUDE CODE ECOSYSTEM		
FOUNDATION	WORKFLOW LAYER	INTEGRATION LAYER
Claude Code (core agent)	Spec-Kit GStack Superpowers	MCP (75+ connectors) LangGraph CrewAI
Claude Agent SDK	cc-sdd Kiro (AWS IDE)	Mastra (TypeScript) Microsoft Agent Fwk

Key 2026 shift: Claude Code is no longer a "chat assistant." It's a full agentic system — it reads files, writes code, runs terminals, calls tools, and can orchestrate other agents. The ecosystem around it is now about **how you structure that agency** — which is exactly where Spec-Kit and GStack come in.

2. What is Spec-Driven Development?

Spec = a contract for how your code should behave, written before a single line of code.

In traditional development:

Idea → Code → (maybe) Docs → Test

In Spec-Driven Development (SDD):

Idea → Spec → Plan → Tasks → AI implements → Review

The spec becomes the **single source of truth**. The AI doesn't guess your intent — it reads the spec and builds to it. This is critical for agents because:

- AI hallucinates without constraints → specs are constraints
- Agents need deterministic checkpoints → specs provide them

3. Spec-Kit (by GitHub)

"Specification as the center of your engineering process"

Open-sourced by GitHub in September 2025. 28K+ stars. MIT licensed.

Core Workflow — 4 Phases

```
/specify → /plan → /tasks → /implement
```

Phase	What happens
/specify	You give a high-level prompt → AI generates a full spec (what & why)
/plan	You give technical direction → AI generates implementation plan
/tasks	Spec is broken into small, reviewable, independently testable units
/implement	AI implements task by task; you review each

4. GStack (by Garry Tan — YC President)

"Turn Claude Code into a Virtual Software Development Team"

Released March 2026. Hit **10,000 stars in 48 hours** — one of the fastest-growing dev tools ever.

Core Concept

Instead of one AI doing everything, GStack assigns **specialist roles** to Claude via slash commands:

<code>/plan-review</code>	→ Engineering Manager
<code>/code-review</code>	→ Senior Engineer
<code>/ship</code>	→ Release Manager
<code>/qa</code>	→ QA Engineer
<code>/browser</code>	→ Browser Automation Specialist
<code>/retro</code>	→ Engineering Retrospective Lead

5. Side-by-Side Comparison

Dimension	Spec-Kit	GStack
Created by	GitHub	Garry Tan (YC)
Core metaphor	Spec as source of truth	Virtual dev team
Entry point	Write a spec first	Start with a plan or code
Primary focus	<i>What</i> to build (specification)	<i>How</i> to ship it (workflow)
Workflow style	Sequential phases with checkpoints	Role-based slash commands
AI autonomy level	Medium (human steers each phase)	High (agent acts like a team)
Best for	New projects, unclear requirements	Active projects, shipping fast

6. When to Use Each

Use Spec-Kit when:

- Starting a **new project** from scratch
- Requirements are ambiguous and need to be clarified first
- You want a **documented spec** as a deliverable (client work, teams)
- Working in a team where multiple people need to understand the plan
- You want to enforce spec compliance across AI model swaps

Use GStack when:

- You already have a codebase and want to **ship faster**
- You need structured **code review and QA** baked into your workflow
- Solo developer trying to simulate team-level rigor
- You're iterating rapidly (startup pace, MVP mode)

7. Roadmap Connection

This maps to **Chapter 16** of the Agent Factory curriculum — *Spec-Driven Development with Claude Code*. The OpenAI Agents SDK skills being built now (agent design, tool use, context management) are the foundation that makes SDD powerful: specs will be used to orchestrate multi-agent pipelines, not just single features.

Sources

- [GitHub Spec Kit Repo](#)
- [GitHub Blog: Spec-Driven Development with AI](#)
- [GStack GitHub \(Garry Tan\)](#)
- [GStack Overview](#)
- [Garry Tan's GStack: Running Claude Like an Engineering Team](#)
- [Spec-Driven Development 2026 Guide](#)
- [Agent Factory — Chapter 16: Spec-Driven Development](#)
- [Superpowers vs GStack Comparison](#)